

JEB160 2024/2025 Sample Final Exam

You have **180** minutes, and you can earn max **60** points. Raise your hand if you think that you found any ambiguity or typo in the text. Put your name on all papers. Note that communication with other students multiplies your points by zero. Good luck!

Hot Hand Fallacy (8 points)

We have a fair coin (i.e., the probability of Head 50%) that shows either Head (H) or Tail (T). We throw the coin 4 times and obtain a string of four outcomes. Suppose we do it many times; eventually, we obtain a distribution of strings, where each unique string occurs with an identical frequency.

Now suppose we test the hot hand hypothesis using a streak of success of length 2 (i.e., HH). This means that we are looking for two events: HHH (after a streak of two successes, HH, another success occurs) and HHT (after a streak of two successes, HH, another success doesn't occur). Specifically, for each string where at least one of these two events occurs, we calculate the *string-specific success rate* (the ratio of HHH events out of both HHH and HHT events). (Strings that don't involve any of these two events are left out as irrelevant.) Then, we calculate the expected/average success rate out of all strings that are relevant, using that each string occurs with an identical frequency.

What is the average success rate? Is it equal, below, or above the true success rate 50%? In other words, is the test biased or unbiased?

Solution

We identify all strings that contain at least one HHH or at least one HHT, and calculate the string-specific success rates:

- HHHH: 2 successes, 0 failures, rate 100%
- HHHT: 1 success, 1 failure, rate 50%
- HHTH: 0 success, 1 failure, rate 0%
- HHTT: 0 success, 1 failure, rate 0%
- THHH: 1 success, 0 failure, rate 100%
- THHT: 0 success, 1 failure, rate 0%

The average success rate is below the shooting success rate, $\frac{250}{600} = \frac{5}{12} < \frac{1}{2}$. As discussed in the class, the procedure is biased; it *underestimates* the actual success rate.

Auction Manipulation (20 points)

Take a standard IPV environment with 2 bidders. Each bidder's value X_i is uniformly drawn from a unit interval, $X_i \sim U[0, 1]$, and the bidders' values are mutually independent, $X_1 \perp\!\!\!\perp X_2$. Each bidder observed her value and is asked to submit a sealed bid for the object. When all bids are collected, the seals are opened, both bids become public, and the object is allocated to the bidder who gives the highest bid. The price is given by is the winner's bid.

Suppose you are Bidder 1 and you know that the default cost is zero; Bidder 2 incorrectly thinks that the default cost is prohibitively large and therefore wrongly concludes that if a bidder wins with the highest bid, she will not default. This superior knowledge gives you an option to manipulate the auction by submitting multiple bids (e.g., through multiple virtual entities) and defaulting (at no cost) down to the price where any additional default would imply that you lose the object. To simplify, suppose you can (at no cost) bid all values on the entire interval (i.e., submit a continuum of bids on the entire interval).

Also, suppose the object cannot be resold so you cannot use the information that you gain by manipulation to earn profits from resale.

1. Characterize your optimal default strategy. That is, if your value is $x \in [0, 1]$ and the opponent's bid (revealed after the bids are unsealed) is $b \in [0, 1]$, explain whether it is optimally to (i) win the object (then characterize also the price that you pay and how exactly you achieve the price by defaulting strategically) or it is optimal to (ii) lose the object.
2. When you apply your optimal default strategy, is the object allocated efficiently or not? (Can we say that the final owner has a higher valuation?)
3. For all pairs (x, b) , characterize the realized profit that you earn when you apply your optimal default strategy.
4. If you are a type $x \leq \frac{1}{2}$, describe your expected profit under the optimal default strategy. (Remember: Bidder 2 doesn't expect you to default; she expects you to behave as a usual bidder in this auction format.)
5. Briefly explain what happens when it is common knowledge that the default cost is zero.

Notice: Use x (instead of x_1) for your value, y (instead of x_2) for the opponent's value and b (instead of b_2) for the opponent's bid.

Solution

1. You can always get the object for a price b by winning with bid 1 and sequentially defaulting down to b . Therefore, the optimal strategy is simple: (i) If $b < x$, you prefer to win the object at the (minimal feasible) price b such that your realized profit is $x - b$. (ii) If $b > x$, you prefer to lose the object.
2. The object is *not always* allocated efficiently. In FPA auction, Bidder 2's strategy is $\beta^{FPA}(y) = \frac{y}{2}$. Therefore, a bid b is made by a type $\hat{y} = 2b$. Three outcomes arise: (i) When $y < x$, the object is allocated efficiently because Bidder 1 prefers to win it ($b = \frac{y}{2} < \frac{x}{2} < x$), and Bidder 1 deserves it ($y < x$). (ii) When $y > 2x$, the object is also allocated efficiently because Bidder 1 prefers to lose it ($b = \frac{y}{2} > \frac{2x}{2} = x$), and Bidder 2 deserves it ($y > 2x > x$). (iii) But when $y \in (x, 2x)$, the object is allocated *inefficiently* because Bidder 1 prefers to win it ($b = \frac{y}{2} < \frac{x}{2} < x$), but Bidder 1 doesn't deserve it ($y > x$).
3. Your realized profit for (x, b) is $\max\{x - b, 0\}$ as if you were bidding truthfully in a SPA. This also demonstrates that the manipulation basically converts the FPA into SPA for the player who manipulates exactly as I mentioned in the class.
4. Bidder 2 uses the optimal FPA bidding strategy, which is $\beta^{FPA}(y) = \frac{y}{2}$. Therefore, a bid b is made by a type $y = 2b$.

Take all winning scenarios, which means to examine all $b \in [0, x]$. (Remember: Your optimal default strategy helps you win and earn profit whenever $b < x$.) This is equivalent to consider the opponent's types $y \in [0, 2x]$. (Notice: If Bidder 2 is of the 'threshold' type $y = 2x$, she bids $b = \beta^{FPA}(y) = \beta^{FPA}(2x) = x$ which makes you exactly indifferent between winning and losing the object; for a higher type, her bid would be too large for you to win the object profitably.)¹

¹Here we can see why we study only $x \leq \frac{1}{2}$. If $x > \frac{1}{2}$, there is no 'threshold' type $\hat{y} = 2x$ at which you are indifferent between winning and not winning; you prefer to win always (even when the opponent is the highest possible, $y = 1$), so the analysis of the winning scenarios changes from $y \in [0, 2x]$ to $y \in [0, 1]$. Generally, we write that your optimal default strategy helps you win and earn profit whenever $y \in [0, \min\{1, 2x\}]$.

- We know that the opponent's type y is distributed by $F(y) = y$, with density $f(y) = 1$.
- In a winning scenario, your realized profit from meeting an opponent with type y is $x - b = x - \beta(y) = x - \frac{y}{2}$.
- Your ex ante expected profit is therefore:

$$\int_0^{2x} [x - \beta(y)] f(y) dy = \int_0^{2x} [x - \frac{y}{2}] dy = F(2x)x - \left[\frac{y^2}{4} \right]_0^{2x}$$

I separated here the benefit side (winning probability $F(2x)$ times the object value x) and the cost side (the expected payment to the seller). After rearranging, the expected profit of Bidder 1 of type $x \leq \frac{1}{2}$ is:

$$2x^2 - x^2 = x^2$$

5. When both bidders know that their default costs are zero, they want to exercise their costless default options which means that they both will submit the continuum of bids and the FPA auction changes into a Dutch auction. We also know that in an IPV environment, Dutch auction is equivalent to FPA. (Precisely, the bidding strategy in FPA is characterized by the sealed bid whereas the bidding strategy in a Dutch auction is characterized by a limit price at which you buy the good. The equivalence means that for a type x , the optimal sealed bid in FPA is identical to the optimal limit price in Dutch auction.) Therefore, what happens is that the manipulator's advantage is eliminated, and we return to an auction that has identical strategic properties; the auction is efficient and equilibrium prices are as if there is no manipulation.

To sum up, in a symmetric IPV environment, it is irrelevant whether the default cost is prohibitive (FPA) or zero (Dutch auction) as long as the level of the default cost is common knowledge.

Note on resale (not part of the required solution)

Since a resale option is not available, you cannot exploit that, when your opponent in FPA submits a bid, you are able to learn her true value and you can resell the object to the opponent. More precisely, the opponent in FPA always bids *below* her true value, and therefore winning the object at the price of her bid (which is always possible due to auction manipulation) and reselling it to the opponent at her value will always give you a

positive profit. (Actually, when the resale option is available and you have full bargaining power, not only that you can gain additional profit from this possibility, but also you are motivated through resale to restore the allocation inefficiency.)

Extension (not part of the required solution)

We can easily illustrate how to generalize the problem to $N > 2$ bidders. (This is not part of the sample exam but shows you how to proceed with a larger N .) The other bidders use the optimal FPA bidding strategy, which is in this IPV environment

$$\beta^{FPA}(y) = \frac{N-1}{N}y.$$

Therefore, if the highest opponent's bid is b , we can find the highest opponent's type $\hat{y}$ by using $\beta^{FPA}(\hat{y}) = \frac{N-1}{N}\hat{y} = b$. In other words, the highest opponent's type that corresponds to the highest opponent's bid b is:

$$\hat{y} = \frac{N-1}{N}b$$

Now, we take all winning scenarios, which means to examine all $b \in [0, x]$. (Your optimal default strategy tells you to win whenever $b < x$.) This is equivalent to considering $y \in [0, \frac{N}{N-1}x]$. Like above, we consider only sufficiently small $x \leq \frac{N-1}{N}$ so that a threshold type $\hat{y}$ exists, $\hat{y} \leq 1$.

- We know that the highest opponent's type y is, in an IPV environment, distributed by $G(y) = F^{N-1}(y)$. Given that $F(x) = x$, we have $G(y) = y^{N-1}$ with corresponding density $g(y) = (N-1)y^{N-2}$.
- In a winning scenario, the realized profit for a type y is $x - b = x - \beta(y) = x - \frac{N-1}{N}y$.
- The ex ante expected profit is therefore:

$$\begin{aligned} \int_0^{\frac{N}{N-1}x} [x - \beta(y)] g(y) dy &= \int_0^{\frac{N}{N-1}x} \left[x - \frac{N-1}{N}y \right] (N-1)y^{N-2} dy = \\ &= G\left(\frac{N}{N-1}x\right)x - \frac{(N-1)^2}{N} \int_0^{\frac{N}{N-1}x} y^{N-1} dy = \left[\frac{N}{N-1}x\right]^{N-1} x - \frac{(N-1)^2}{N} \left[\frac{y^N}{N}\right]_0^{\frac{N}{N-1}x} = \\ &= \left[\frac{N}{N-1}\right]^{N-1} x^N - \left[\frac{N}{N-1}\right]^{N-2} x^N = \frac{1}{N-1} \left[\frac{N}{N-1}\right]^{N-2} x^N \end{aligned}$$

Two Prizes Instead of One (12 points)

There are $n = 3$ contestants competing over a divisible prize $R = 3$. The prize is allocated in an all-pay auction.

1. What is the optimal distribution of investments, denoted $F(x)$, that each contestant uses in the equilibrium? Derive mean, median and maximal investment of each contestant.
2. Now suppose that the prize is divided into two prizes 2 and 1. Each contest can win at most one prize. The contestant with the highest investment gets the prize 2 and the contestant with the second highest investment gets the prize 1. What is the optimal distribution of investments, denoted $H(x)$, that each contestant uses in the equilibrium? Derive mean, median and maximal investment of each contestant and compare with the case when only one prize was available.
3. We can see that $F(x)$ and $H(x)$ are different with the exception of two specific levels of investments: The first exception is obviously the minimal investment, since at $x = 0$, $F(0) = H(0)$. What is the second investment level at which $F(x) = H(x)$? In other words, at which positive investment level is the probability of winning the highest prize (in the case with a single prize, this is actually the only prize) identical in both contests?

Solution

1. Single prize: This problem was solved in the class, $F(x) = \left(\frac{x}{R}\right)^{1/(n-1)} = \left(\frac{x}{3}\right)^{1/2}$.
 - The median $\tilde{x}$ is defined as $F(\tilde{x}) = \frac{1}{2}$, which gives $\tilde{x} = \frac{3}{4}$.
 - The maximum $\bar{x}$ is defined as $F(\bar{x}) = 1$, which gives $\bar{x} = 3$.
 - The mean $\mathbb{E}(x)$ can be solved from equalizing property, whereby the ex ante expected profit is zero, $\frac{R}{n} - \mathbb{E}(x) = 0$, and therefore $\mathbb{E}(x) = 1$. Alternatively, you may solve for $\mathbb{E}(x) = \int_0^3 xf(x)dx$, where $f(x) = \frac{1}{2} \left(\frac{1}{3}\right)^{1/2} x^{-1/2}$.
2. Two prizes: In the class, we illustrated the difference by comparing the course evaluations at FSV and IES. In the specific example analyzed here, we proceed as follows:
 - As usual, we analyze a symmetric equilibrium where each contestant uses a distribution $H(x)$.

- With an investment x , the probability of winning the higher prize 2 is $H^2(x)$; the expected benefit is therefore $2H^2(x)$.
- With an investment x , the probability of winning the lower prize 1 is $2H(x)[1 - H(x)]$; the expected benefit is identical, $2H(x)[1 - H(x)]$. (Winning the lower prize happens in two types of scenarios. For example, suppose I am player 1. The first scenario is $x_2 < x_1 < x_3$ and the second scenario is $x_3 < x_1 < x_2$. Each scenario has probability $H(x_1)[1 - H(x_1)]$.)

The two expected benefits can be summed up because the two events never occur jointly. This gives that the expected payoff for an investment x is $2H(x) - x$. By equalizing property (and using that with zero investment, the benefit is zero and therefore the expected payoff is zero), we have $H(x) = \frac{x}{2}$.

- The median is $\tilde{x} = 1$ which is *larger* than with a single prize. (Dividing the prize into multiple prizes reduced incidence of very small investments.)
 - The maximum is $\bar{x} = 2$. (Dividing the prize into multiple prizes discouraged making very large investments.)
 - The mean $\mathbb{E}(x)$ can be again solved from equalizing property, i.e., $\mathbb{E}(x) = 1$. Alternatively, you may solve for $\mathbb{E}(x) = \int_0^2 xh(x)dx$, where $h(x) = \frac{1}{2}$.
3. Trivially, we are solving $F(x) = \left(\frac{x}{3}\right)^{1/2} = \frac{x}{2} = H(x)$, which gives (for $x \neq 0$) a cutoff which is above the mean,

$$x = \frac{2}{\sqrt{3}} \doteq 1.15 > 1.$$

For any lower x , we observe $F(x) > H(x)$ (competition over a single prize encourages very low investments more frequently); for any higher x , we observe $H(x) > F(x)$ which is equivalent to $1 - F(x) > 1 - H(x)$ (competition over a single prize encourages very high investments more frequently).

Miscellaneous (20 points)

1. When we introduced revenue equivalence, we said that an auction is either standard or not standard.
 - (a) What characterizes whether an auction is standard or not?
 - (b) Is there a contest which is also a standard auction? (That is, characterize the contest success function for which the contest is a standard auction.)

- (c) Which contests are not standard auction? (That is, characterize a property of the contest success function under which the contest is a not standard auction.)

Solution:

- (a) An auction is standard if the highest bid wins the object. An auction is not standard if it may happen that the highest bid doesn't win the object.
 - (b) The property above means that the prize is allocated through a perfectly discriminative contest. In other words, only all-pay auction is both a contest and a standard auction.
 - (c) All other (less than perfectly discriminative) contests are not standard auctions.
2. When is an auction (or contest) a game with complete information? And when is it a game with incomplete information?

Solution: The game is with complete information when all components of the game (players, strategies, outcomes, mappings from strategies to outcomes and preferences over outcomes) are common knowledge. The game is with incomplete information when all components *except for preferences over outcomes* are common knowledge.

In the context of auctions, preferences over outcomes are represented by bidders' values. Therefore, an auction is a game with complete information when bidders' values of the object are common knowledge and incomplete information when bidders' values of the object are not common knowledge. (This is because the values are drawn from some joint distribution and each bidder observes a specific private signal/own type.)

In the context of contests, preferences over outcomes are represented by the players' values of the prize. So existence of common knowledge about players' value of the prize characterizes whether the contest is a game with complete or incomplete information.

3. Does revenue equivalence apply also to auctions with known values? To answer that, recall our analysis of the sealed-bid auctions (FPA and SPA) with known values $v_1 \gg v_2$ (both integers) where the bidders submit only discrete bids (integers).
- (a) Using the Nash equilibrium as a solution concept, can we argue that the two auctions are revenue equivalent?

- (b) Using the Nash equilibrium in weakly undominated strategies as a solution concept, can we argue that the two auctions are revenue equivalent?

Solution: In this environment, the set of Nash equilibria is different in FPA. In FPA, Nash equilibria are along the diagonal where the price (winning bid) is in the interval $[v_2, v_1]$. In SPA, the set of Nash equilibria is larger, but the price (second bid) is in the interval $[0, v_1]$. Using NE only, SPA gives a richer set of predictions on revenues, so we conclude that the auctions are not revenue-equivalent.

However, adding the criterion of weak dominance, there is a *unique* profile in weakly undominated strategies in both auctions. In FPA, it is $(b_1, b_2) = (v_2, v_2 - 1)$ and it leads to the price v_2 . In SPA, it is $(b_1, b_2) = (v_1, v_2)$, and it leads to the price v_2 . The auctions are then revenue-equivalent.

4. Unexpected revision of the reserve price: Suppose that the seller organizes SPA with a reserve price in an IPV environment. It is expected by the seller as well as bidders that, if the auction fails, there is no other selling opportunity. Now, suppose that the auction actually fails (no bidder is willing to pay at least the reserve price), but there *unexpectedly* emerges an opportunity to run one extra SPA auction with a reserve price. In this second (unexpected) auction, how does the seller revise the reserve price? Will he decrease it or? Will he set the reserve price to her own value x_0 or not?

Solution: Since the existence of another selling opportunity was unexpected for both the seller and bidders, the seller set the reserve price optimally as $r > x_0$, and the bidders used ‘truthful’ bids (i.e., didn’t participate if $x < r$ and participated if $x \geq r$).

The auction failure thus gives information to *all* participants (seller and bidder) that the players’ values are drawn not from i.i.d. distributions $F(x)$, but from truncated distributions $F^r(x) = \frac{1}{F(r)}F(x)$. (In other words, the initial unconditional distribution was $X \sim F$, whereas the conditional distribution $X \mid X < r \sim F^r$.)

From the perspective of the seller, a bidder may be either unprofitable ($x \leq x_0$) or profitable ($x_0 < x < r$). This means that the seller wants to take this extra opportunity to try to sell the good to the profitable bidders. When solving her new optimal reserve price r' , it is clear that

$$x_0 < r' < r.$$

The first inequality follows from the analysis in the class (the problem is the same, only F is replaced by F^r), and the second inequality is trivial (if not, then the bidders again wouldn't participate because in the initial auction, their participation decision was based on their true value).

5. The Czech Treasury (Ministry of Finance) calls the discriminatory-price multi-unit auction of Treasury Bills as 'American' auction, referring to adoption of this format by the U.S. Treasury in the late 1990s. It calls the uniform-price multi-unit auction of Treasury Bills as a 'Dutch' auction. Now, we know that in the auction practice, a 'Dutch' auction represents a completely different auction format (an open descending price auction for a single object). Using knowledge about similarities and dissimilarities of auctions that you have learned, is using 'Dutch' as an adjective for these two different auctions a bit or small problem? Discuss.

Solution: We know that Dutch auction (for a single object) is strategically equivalent to a sealed bid FPA (for a single object). We also know that there is similarity between sealed bid FPA (for a single object) and *discriminatory-price* multi-unit auction (recall analysis in Kastl, 2020). Therefore, it is truly misleading and confusing to use 'Dutch' for a uniform-price multi-unit auction.